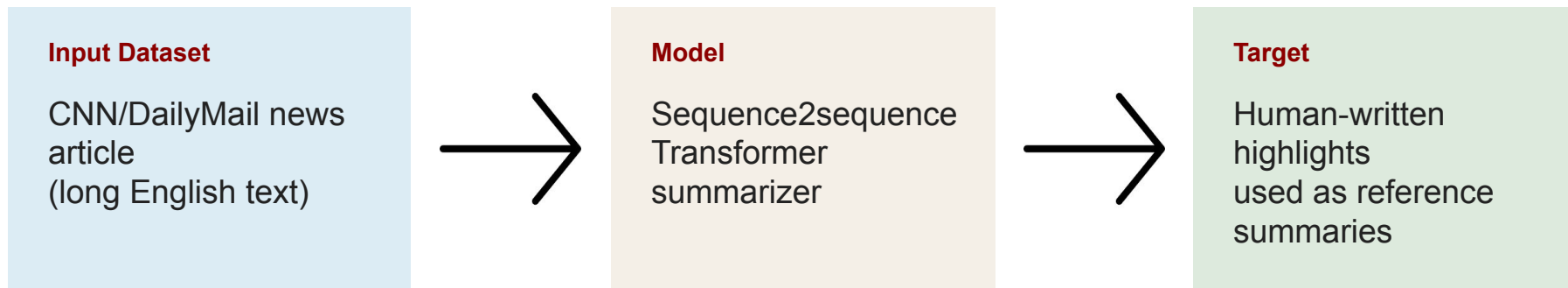

Abstractive News Summarization with Transformer Models

Yihong Li, Tony Li



Problem & Dataset

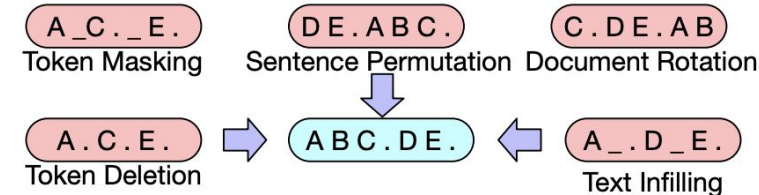
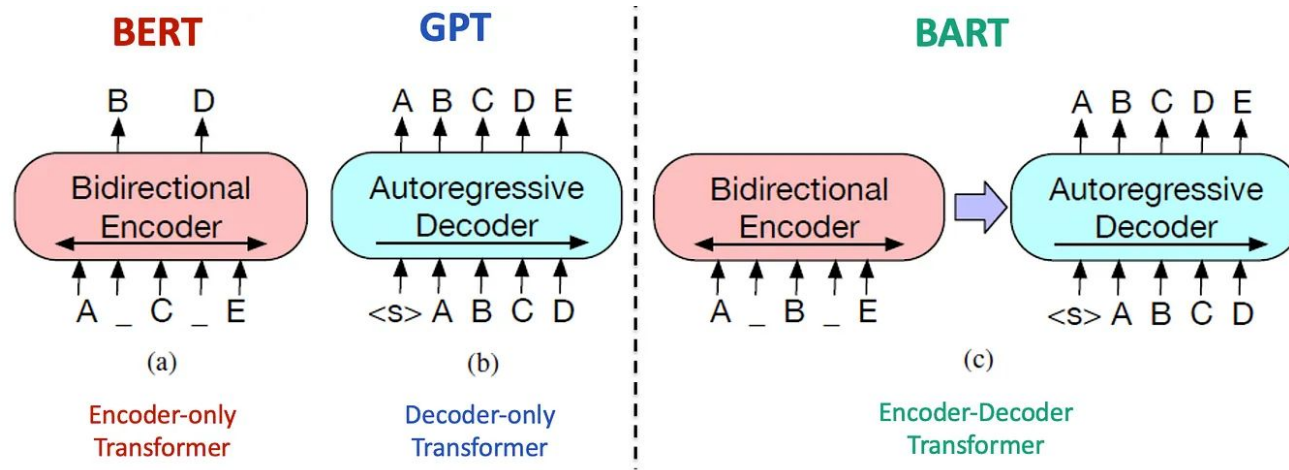
Task: generate concise abstractive summaries for long-form news articles.



Dataset field	Role	Description
article	Source input	Full news article to summarize
highlights	Target output	Journalist-written bullet-style summary
official splits	Evaluation structure	Train / validation / test from CNN/DailyMail

Why BART for Summarization?

BART(Bidirectional and Auto-Regressive Transformers) keeps the encoder-decoder Transformer backbone, but pretrains it as a denoising autoencoder.



Systems Compared

We compare our fine-tuned abstractive model against a simple extractive baseline and a stronger reported reference.

System	Type	Role in project
Lead-3	Extractive baseline	Uses the first three article sentences; simple but strong for news.
Fine-tuned bart-base	Our main model (140M)	BART-base trained by us on CNN/DailyMail subsets.
bart-large-cnn	Reference only (406M)	Already fine-tuned on CNN/DailyMail, so not counted as our main training result.

Why Lead-3 matters: news often follows an inverted-pyramid structure, so the opening sentences can be surprisingly competitive.

Training Setup

Choice	Setting used in our experiments
Main model	facebook/bart-base
Training sizes	20k and 50k CNN/DailyMail training examples
Sequence lengths	1024 input tokens, 128 target tokens
Hardware	Colab A100 GPU with BF16 mixed precision
Optimization	2 epochs, learning rate 3e-5, dynamic padding
Automatic metrics	ROUGE-1 / ROUGE-2 / ROUGE-L / ROUGE-Lsum and BERTScore

Implementation detail: dynamic padding avoids wasting memory on shorter articles, while 1024 input tokens uses the maximum standard BART context window.

Evaluation Snapshot

Fine-tuned BART-base improves over Lead-3; more training data gives a modest gain.

ROUGE-Lsum

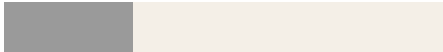
Lead-3 baseline  **36.46**

BART-base 20k  **38.07**

BART-base 50k  **38.36**

BART-large-CNN ref.  **40.04**

BERTScore F1

Lead-3  **24.39**

BART 20k  **31.67**

BART 50k  **32.10**

BART-base beats the Lead-3 baseline on both metrics. The 50k training run is also slightly better than the 20k run, so more training data helps, but the improvement is modest. The BART-large-CNN reference is still stronger on ROUGE-Lsum. This is expected because it is a larger model and already fine-tuned on CNN/DailyMail.

Key Takeaways

Fine-tuning works

BART-base beats the Lead-3 extractive baseline on both ROUGE and BERTScore.

More data helps, but modestly

The 50k run improves over the 20k run, but gains are not dramatic.

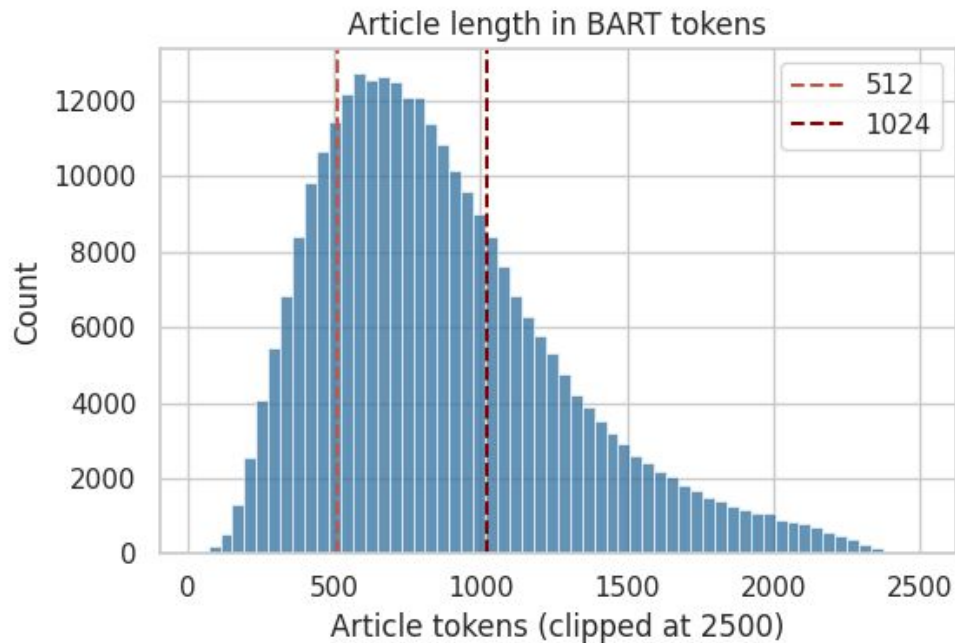
Reference model remains stronger

BART-large-CNN is larger and already trained on CNN/DailyMail, so this gap is expected.

Next question

How much do context limits and long-article truncation affect summary quality?

Truncation Problem



79.54%

cleaned train articles exceed 512
BART tokens

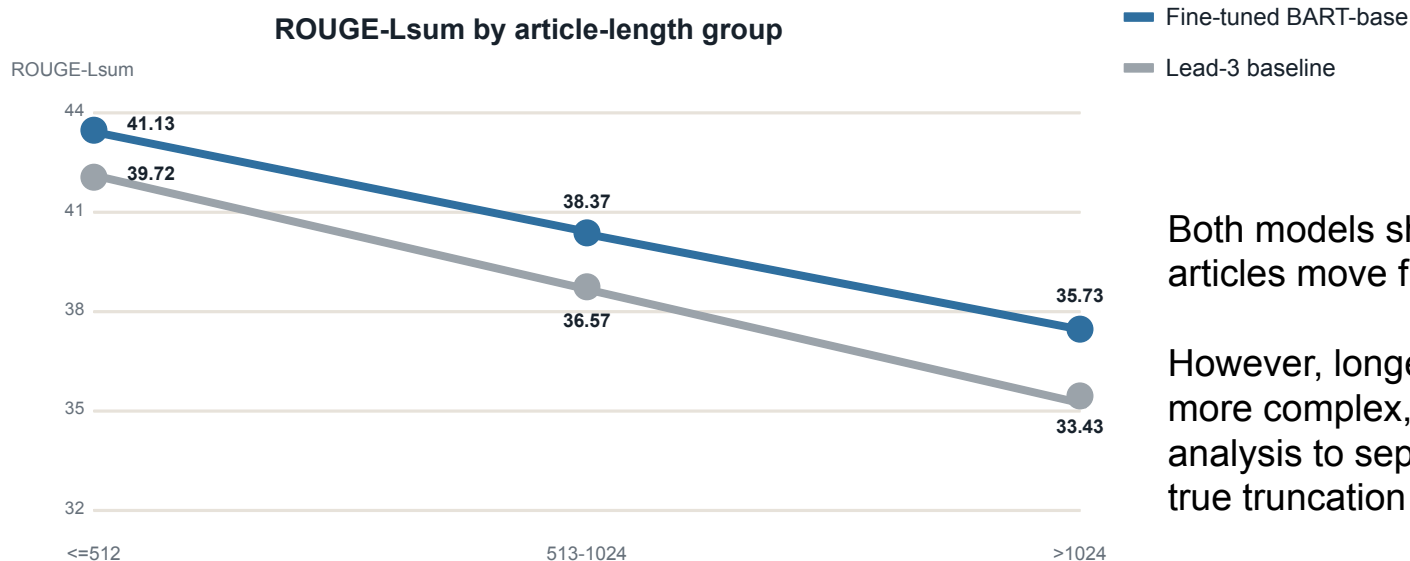
30.04%

cleaned train articles exceed 1024
BART tokens

Split	<=512	513-1024	>1024
Train	20.46%	49.51%	30.04%
Validation	24.24%	46.99%	28.77%
Test	23.74%	47.04%	29.22%

Article Length vs. Summary Quality

Summary quality decreases as article length increases, which suggests that long inputs are harder for the model. However, this pattern alone does not prove that truncation is the direct cause.



Both models show lower scores as articles move from short to long groups.

However, longer articles may also be more complex, so we need further analysis to separate input length from true truncation effects.

News Lead Bias

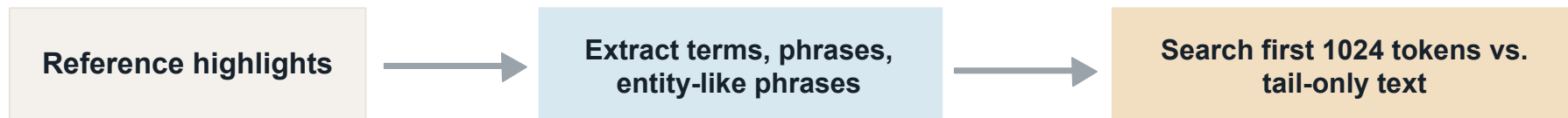
- CNN articles often follow an inverted-pyramid structure, where the main event and key facts appear near the beginning.
- The first 1024 tokens may already contain much of the information needed for a reference-style summary.

Inverted-pyramid news structure



Reference Coverage

Most reference-summary content is already covered by the first 1024 tokens, while the article tail adds limited unique information.



Coverage measure	Mean	Median
Prefix term coverage	79.63%	81.74%
Tail-only terms	4.06%	0.00%
Prefix entity coverage	87.12%	91.49%
Tail-only entities	2.27%	0.00%

Prefix coverage is high for both terms and entity-like phrases.

Tail-only reference information is low, especially for entities.

Long-Context Experiment Design

Hierarchical BART: Splits the long article into chunks, summarizes each chunk, and then combines the chunk summaries into one final summary.

Fine-tuned LED: Uses a long-context encoder-decoder model that can directly process longer inputs than BART, up to 4096 tokens in our experiment.

Truncated BART

first 1024 tokens -> summary

Hierarchical BART

chunks -> chunk summaries -> final summary

Fine-tuned LED

long-context encoder-decoder, up to 4096 tokens

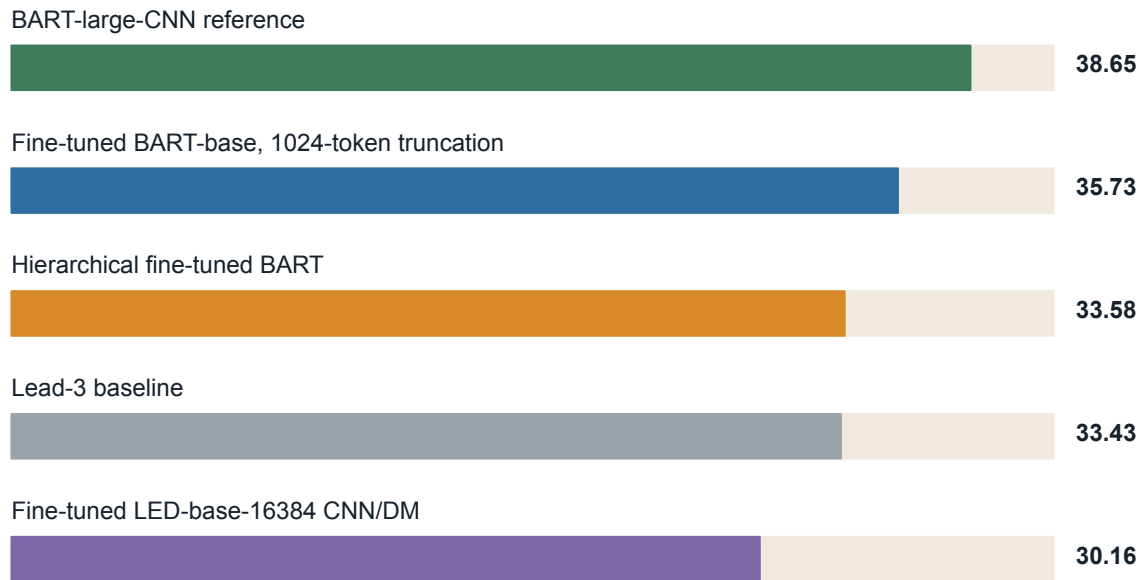
BART-large-CNN

strong CNN/DailyMail reference model

Main Finding

Longer context did not improve performance -> Most useful information was already in the prefix

Long-context methods introduced extra weaknesses: Hierarchical BART may lose details during compression, while LED may not use the extra context effectively.



Qualitative Evaluation

Method	Fluency	Factual	Coverage	Concise	Overall
BART-large-CNN	4.87	4.97	3.57	3.93	4.33
Lead-3	4.80	5.00	2.50	3.40	3.93
Fine-tuned BART	4.37	4.77	2.80	3.37	3.83
Hierarchical BART	4.27	4.63	2.50	3.60	3.75
LED	3.20	3.80	2.27	3.03	3.08

Method-level error pattern

- **Lead-3:** Very factually safe because it copies the opening, but it often misses reference-relevant details later in the article.
- **Fine-tuned BART:** Improves coverage over Lead-3, but still loses points for omissions and occasional unsupported or verbose details.
- **Hierarchical BART:** Accesses more text, yet chunk-level compression does not translate into better coverage or overall quality.
- **LED:** Longer input access is not enough here; lower factual and fluency scores suggest style mismatch and more unstable generation.

Real-News Demo

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Generated summary

SpaceX's regulatory filing revealed a financially smart link between its launch-services division and profitable Starlink satellite internet operation. Its AI business looks shakier than its rockets. Revenue in the AI division has mostly come from X, which is not a pure-play AI venture.